

Exercise 1: Treatment Design

You are planning an online experiment to test different design variations of your online store, seeking to learn which of them is most efficient to engage visitors on the site. As a first variable, you have come up with 4 different color schemes. As a second variable, you have created 3 different text introductions that greet the website's visitors.

- a) Sketch out a treatment matrix and discuss your major design choices for the experiment – including within/between subjects design, sample size, subject-to-treatment assignment, main outcome variable, etc.

A-B Test

<i>Treatment</i>	<i>Color 1</i>	<i>Color 2</i>	<i>Color 3</i>	<i>Color 4</i>
<i>Text 1</i>	...	...	...	...
<i>Text 2</i>	...	...	...	...
<i>Text 3</i>	...	...	...	...

Each user visiting the online store is shown exactly one color-text combination of the website (full-factorial, between-subject design). Per cell, at least 30 visitors are needed to draw meaningful conclusions, in practice likely much more visitors are available per cell. The IP address is a good tool for subject-to-treatment assignment, i.e. subjects are evenly distributed towards treatments based on their unique IP address. Different main outcome variables could be of interest, i.e. revenue, time spent on certain sites, clicks, conversion rate, newsletter signups to name a few.

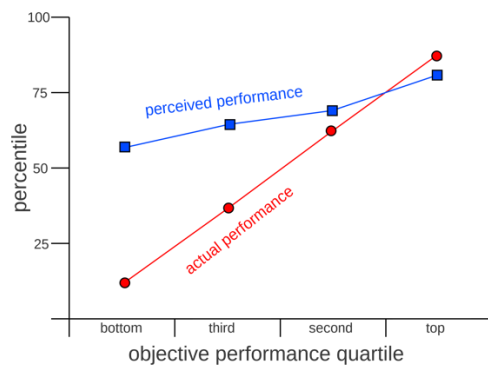
Exercise 2: Dunning-Kruger Effect

You may have heard about the Dunning-Kruger effect (if not, make yourself familiar with it by a brief Internet research). In particular, you want to know whether telling people about this effect will reduce (or even increase it). In your own words, sketch out an experimental design to test whether this effect actually exists. Please note that you will have to recruit subjects from a University-based subject pool and will conduct the experiment at your institution's economics laboratory. Your research budget for paying off participants is 800€.

Be as specific as possible with regard to the following questions:

a) What is the Dunning-Kruger effect?

The Dunning-Kruger effect is a hypothetical cognitive bias stating that people with low ability at a task overestimate their ability (especially/ more than others). See Illustration below:



b) What are your main research questions?

What is the behavioral effect of telling people about the Dunning-Kruger effect (on the effect itself)?

c) How does your treatment design look like?

It is a between-subjects design with the two-treatment conditions TELL and NOTELL. Subjects are randomly assigned to either one of these conditions (ideally, a 50:50 split). In the TELL condition, subjects receive the information on the Dunning-Kruger effect, in the other condition, they do not.

d) Which experimental task would you choose?

The task could be to click through 100 images of international public persons (from sports, politics, history, film, entertainment, literature, ...) and type their name (allowing for some slack). The set of images is the same for all participants. Sequence of images is randomized. Subjects have a total of 15 minutes = 900 seconds = 9 seconds per image.

Before starting the task, subjects provide a self-assessment of how good they believe to be at the task (i.e., which percentage they will identify correctly 0-100).

Alternatively, one could choose simple math tasks (i.e., multiplication of single- and double-digit numbers). However, this has the drawback that performance would also be contingent of effort, not only ability.

e) How do you incentivize participants?

We can assume a duration of approx. 45 minutes for preparation, execution and debriefing of the experiment. An average payment of €12 (i.e., €16/hr) would be adequate. The self-assessment is incentivized in that each %-point deviation from their actual score (above OR below) will cost them 1 MU. However, each correctly identified

person will win them 3 MU. Thus, there is no incentive to perform badly on purpose if they realize that they have underestimated their ability.

Average performance on the data set is pre-tested and adjusted accordingly so that overall, a performance of ~75% can be expected ($75 \times 3 \text{ MU} = 225 \text{ MU}$). Assuming that the average difference between estimated and actual performance is 25, average subjects would walk away with 200 MU.

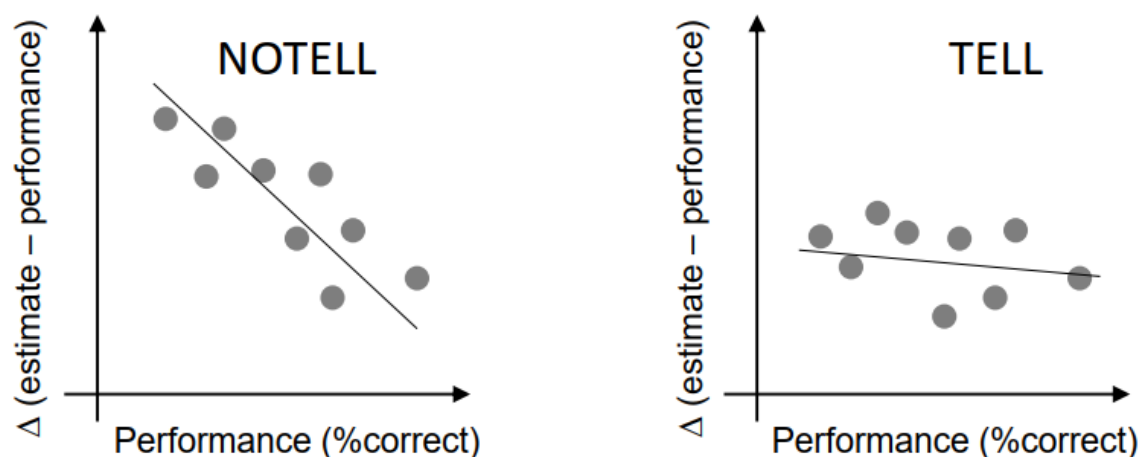
We assume a fixed payment component of 5€. In order to hit an average payoff of €12 in total, 1 MU should be set to $7.00 \text{ €} / 200 = 0.035 \text{ €}$.

Maximum outcome: estimate 100% and then actually perform 100%: $5\text{€} + 3 \times 100 \times 0.035\text{€} = 15.50\text{€}$.

Minimum outcome: estimate 100 and perform 0: $5\text{€} - 1 \times 100 \times 0.035 = 1.50\text{€}$

f) What is the experiment's main outcome variable?

%-point deviation of self-assessment from actual score or, more precisely, difference between the two regression slopes:



g) How large should you sample be and how do you recruit participants?

Participants are invited via email and show up at the lab. Sample size could be around $800\text{€} / 12 \text{ € (average payout)} \rightarrow \text{around } 66 \text{ participants}$

h) Which ethical considerations would you make?

Participation is voluntarily. Using university-based subjects pools makes sure that subjects actually want to take part in experiments.

Target group is not particularly vulnerable

Experiment procedure is communicated to subjects adequately.

Participants can withdraw from the experiment at any time.

Subjects are compensated adequately for taking part.

Personal data should be kept private. No exposure of individual performance/payoffs to other subjects.

Risks of partaking are rather low. There is no physical risk. There are no disturbing or otherwise distressful materials. There is a rather limited amount of time pressure (no more severe than students would face in exams regularly). Debrief after experiment completion should be provided.

Exercise 3: Peak-end rule

You may have heard to the *peak-end rule* (if not, look it up :). Your task now is to devise an experiment that tests this theory's claims in a clear-cut manner. **Hint:** You will need to think of a specific task, a treatment design, a suitable sample, the precise procedure, stimulus materials, incentive structure, and measurement variables – among other things.

Basic idea: 5 videos of stand-up comedy, each 30 seconds long. Each participant will have to rate how funny they found the videos. The videos are pre-tested with a test audience to collect average scores (e.g., on an ascending scale from 1-10)

Clip	1	2	3	4	5
Duration	30 sec	30 sec	30 sec	30 sec	30 sec
Score	4.7	8.5	2.1	6.9	5.9

Now, we randomly show 3 out of 5 videos to each of our participants. After each sequence of videos, the participants are asked to rate how funny they found the videos overall. As we select 3 out of 5 videos, we can pick 10 different combinations of videos. As we randomize the order of videos, we get 6 different ways to do this for a total of 60 different treatments.

$$\# \text{ of sequences} = \frac{n!}{(n-k)!} = \frac{5!}{(5-3)!} = \frac{5 * 4 * 3 * 2 * 1}{2 * 1} = 5 * 4 * 3 = 60$$

Because we know both the video with the highest score per sequence as well as the score of the last video in the sequence, we can check for the peak-end rule (of course, we have to also check for other theories / explanations).

Sequence	Treatment	$\emptyset$	Peak	End	Participant rating
1	1 -> 2 -> 3	5.1	8.5	2.1	?
2	1 -> 3 -> 2	5.1	8.5	8.5	?
3	2 -> 1 -> 3	5.1	8.5	2.1	?
...	...	...	...	...	...
60	...	...	...	...	...

Exercise 4: Field Experiments

Given that field experiments offer many advantages, in particular regarding external validity, why should researchers not always conduct field experiments?

a) Definitions

Laboratory experiment: A study conducted in a controlled, artificial environment (typically a dedicated lab with cubicles and computers), where participants are recruited from a subject pool, know they are part of an experiment, receive explicit instructions, and are paid through a structured, behavior-dependent incentive scheme. The researcher controls the decision context, the rules, the information available, and the payoff structure.

Field experiment: A study conducted in a real-world setting (a marketplace, a platform, an organization, a public space), where participants typically do not know they are part of an experiment, face natural payoff structures, and act under conditions the researcher can only partially control. Treatment is delivered by manipulating something in the real environment (e.g., a profile photo, a rating, a price tag).

b) Why not always go to the field?

Field experiments tend to score high on external validity, but they are not always the right, or even a feasible, choice.

Access and feasibility (the main point). The binding constraint is usually access rather than money. Field experiments can even be cheaper than lab studies, but you typically need the right environment and a cooperating partner: Amazon to run an A/B test on its site, eBay to vary an auction format in country X, a city authority to change a traffic-light schedule, or simply a public street to run a charity-solicitation study with mixed-gender teams. Without that access, the study often cannot happen, regardless of budget. Audit studies (e.g., posting X listings on Airbnb) work for a similar reason: the platform is open enough to enter. Lab procedures, by contrast, are documented, reproducible, and don't depend on a third party.

Internal validity and control. In the lab, the researcher can hold almost everything constant except the manipulated variable (*ceteris paribus*). In the field, many confounds such as weather, time of day, prior interactions, algorithm changes, or news events cannot easily be observed or controlled, so clean causal claims are harder to defend. This also covers two related issues: measurement (lab data capture every click, decision, and response time, while field data are typically just an outcome like booking yes/no) and incentives (the lab implements the exact payoff structure the researcher intends via Induced Value Theory, whereas in the field actual valuations are heterogeneous and largely unobserved).

Manipulation of the variable of interest. Many variables are difficult or impossible to vary in the field. You generally cannot redesign eBay's auction format, change Airbnb's review system, or alter tax rates on a whim. Some of this becomes possible with a cooperating partner, but many manipulations (e.g., disabling consumer protections)

would likely not be permitted on an official platform. The lab lets you implement institutions that don't (yet) exist and compare them head-to-head.

Ethics and consent. Field experiments often involve subjects who don't know they are participating, with the hosts in Edelman et al. (2017) being a classic case. Audit studies in particular can rely on deception (e.g., fake job applications or fake bookings), which is hard to reconcile with informed consent, data protection, and protection of uninvolved third parties. Lab participants volunteer, are briefed, can withdraw, and are debriefed.

c) When is each appropriate?

A reasonable default is: use the field whenever you can. The lab almost always works, since you can recruit subjects and run a session. But the field doesn't always work, because access, cooperation, and feasibility are real constraints (see b). That makes field opportunities valuable whenever they arise.

That said, the two methods are largely complementary rather than substitutes:

Use the lab when the goal is to test a theory cleanly, isolate a mechanism, or study an institution that doesn't exist yet. The lab is also where you can more easily capture moderators and mediators, that is, the reasons why an effect occurs, since you control the environment and can measure the process and not just the outcome.

Use the field when external validity is paramount: testing whether a lab effect generalizes to real markets, estimating effect sizes for policy, or studying behavior the lab cannot easily produce in an authentic way (such as long-term trust or real reputational stakes).

A caveat on leaning too heavily on A/B testing. Pure A/B tests in the field tend to tell you what works, for instance which layout gets more clicks or which price converts better, but often not why. Amazon doesn't necessarily need to know the mechanism, since whichever variant wins, wins. For research, that can be a limitation: you may learn less about the underlying theory, and the result is more mechanistic than explanatory. There is also an ethical consideration: when Facebook A/B-tests a feature on a million users, those users are influenced either way without meaningful consent, which is a grey zone that lab experiments, with their explicit briefing and debriefing, avoid.

Strong research programs therefore often run a phenomenon through both: lab to establish the mechanism, field to show it survives outside the lab.